

ActInf ModelStream #002.1 ~ Noor Sajid & Philip Ball

ModelStream | Noor Sajid & Philip Ball | 1:15:43

hello

and welcome everyone to the active

inference lab

this is the model stream number 2.1

on april 16 2021

and today is going to be an awesome

model stream we're just going to

briefly go around and introduce

ourselves and then

i'll just mention how the session will

be run today and then we'll pass it to

noor for a presentation so i'm daniel

and i'm a postdoctoral researcher in

california i'll pass to

philip

hi yeah i'm currently a phd student at

oxford

in my second year and yeah i guess this

is work i did kind of before i started

my phd

um alongside nor and i've currently more

focusing on

data efficiency within specifically

reinforcement learning but

i guess on that topic like a knowledge

of active inference

is obviously useful for approaching such

research problems

who and or hi

hi i'm noah i'm a third year phd student

in the theoretical neurobiology group at

the welcome center for human

neuroimaging

um at ucl um so that's university

college london

um so my phd supervised by cars focused
on these ideas
pertaining to adaptation one of which
i'll be focusing on today which is
behavioral adaptation in non-stationary
environments using active inference
so thank you awesome thanks both for
joining and for this presentation we're
going to be
hearing a who knows how long
presentation from nor
and then i'm gonna be compiling
questions from the chat
so please just type questions as they
come to you
and then we'll address them at the end
so thanks again and nor please
take it away perfect thank you
um so today i'll be presenting some work
um
that i did in collaboration with philip
who you've just heard from
thomas parr and carl fristan um
so it's titled active inference uh
demystified and compared
okay okay perfect okay so the
presentation structured as follows
can you hear the screens
i think it's sharing or was it are you
not able to see it i'm not seeing it
could you just re-share it
yep sure
um technology i tell you
there we go and i'll crop it so go for
it thanks
perfect thank you so the presentation
structured as well as
first i'll briefly motivate the problem

setting and provide
details of a particular active inference
instantiation under consideration today
which is the discrete state space
setting
and the second half of the presentation
is going to be focused on some
particular examples comparing
the active influence formulation with
reinforcement learning
specifically q learning and bayesian
model based
algorithm and then what i'm going to do
is provide some face validity of
particular aspects of why you would want
to even use active inference
okay um so what is active inference
um it's a first principles account of
how biological
or artificial agents may operate in
dynamic non-stationary settings
it stipulates that these agents in order
to maintain homeostasis
reside in attracting states that
minimize their entropy or their surprise
so if you take this particular example
that we're seeing of this past
little hungry agent opening the fridge
the way it would work is like you would
need to why would you open the fridge
right so you want to make a particular
choice between
eating at home or outside and in order
to do that you
have to decide what is the optimal
action that would allow you to resolve
your own uncertainty about the current
stage of affairs

so um and that would help you then
decide whether you want to cook at home
or you want to walk to the restaurant
and this particular instance this has
led to the agent opening the fridge to
check whether it even has food at home
and what's nice about active inference
is that it allows you to
think about these problem settings in a
more formal way by specifying that
optimal behavior
is determined by evaluating the evidence
that is the sensory input
under the agent's generative model of
observations that it
is being exposed to and in this
particular
presentation what we'll do is focus on
just the process theory that underwrites
active inference and not talk through
the biological
on the neural plausibility of the active
inference message passing scheme
um but to properly uh motivate why we
would want to even use
um active inference in comparison to
generic reinforcement learning
algorithms
we need to first start off with this
understanding that with an active
inference there's this commitment to a
pure belief based scheme
which means that reward functions are
not always necessarily um
sorry unnecessary because any policy
that you would have
has epistemic value even in the absence
of preferences

additionally active inference agents can also learn their own reward functions and this helps the agent describe the type of behavioral that it expects to see in itself as opposed to something that it would get from the environment and these two particular points are really important in contrast to reinforcement learning because under standard RL settings the reward function would define how the agent interacts or behaves within a particular environment setting but defining that reward function in the first place is quite difficult because it assumes that there is a specific signal that's being given from the environment that can be unambiguously good or bad for the agent which wouldn't necessarily hold true in a real setting where these environment signals can change depending on the setting for example eating ice cream is not always going to be rewarding if you're ill and it might make you worse and that's why constructing these reward functions in the first place is extremely difficult and if you're not constructing them in an appropriate way even with an RL setting it can result in sub-optimal behavior for your agents

um an active inference is really good in the sense because we are actually

um replacing or bypassing the traditional reward function that you would have in the rl setting with prior beliefs about preferred outcomes so the sort of desired states of affairs that you want to see yourself be in and this becomes important in settings where there's no reward or there's really imprecise understanding of what a reward or preference setting should look like and in this scenario within the standard active inference discrete state formulation what we can do is learn the empirical prior distribution over these preferred outcomes um and intrinsic uh sorry the internal reward function of that agent and this sort of brings me to like the first distinct um conceptualization between rl setting and active inference because within active inference rewards are nothing distinct they're just a standard observation that the agent is getting from the environment whereas in rl they're quite necessary to have appropriate action um that the agent is going to learn um the second point i we want to make sorry we wanted to make was that active inference provides a principal account of epistemic exploration and intrinsic motivation as minimizing uncertainty

and again within the rl setting this is quite crucial because the whole uh premise of like lots of new algorithms that we see in our nrl is to try and find the right trade-off for the balance between exploration and exploration so what are the right set of actions that the agent should make at a given point in time should i carry on choosing all the different ice cream flavors that it's been never been exposed to um like mustard etc or should it always use this oh sorry always have the same ice cream flavor that's been exposed to in the past that it really likes like for example hazelnut or nutella for instance um so this um is it's an outstanding problem with an rl and under a bayesian framework active inference deals with it naturally um using um the expected free energy formulation that i'll come to in a moment um and the last bit that you can see within the active inference framework is that it naturally accounts for uncertainty as part of the belief updating process okay so now that i've sort of um laid out the three things that are super interesting about the active influence scheme in

comparison to rl i'm going to
provide some intuitions as to why um
you can sorry some motivations as to
why we can even formulate the active
influence formulation the way we do
okay so um
sorry i just realized i like skipped
ahead um
no no sorry great presentation
yep thank you so much and let me just
scroll down to um
okay so i previously stated that with an
active inference
it stipulates that agents are
maintaining their homostasis
by residing in attracting states that
minimize surprise
so you must have been thinking what is
surprise well um
how we define surprise as a negative log
probability of outcomes
and for this we introduce one random
variable which is
o that corresponds to a particular
outcome that's received by an
agent and this um exists within a finite
set of
all possible outcomes and that's o here
so um
first equation that we have just um
formally states that out
and here p denotes the probability
distribution over
outcomes
um okay so in active inference the way
the agent will actually
minimize this surprise quantity that we
just walked through is by maintaining a

gender model of the world
um and this is important because at any
given point in time the agent wouldn't
necessarily have access to
the true measurements of the current
state of the world so in this particular
graphic that you see here
you've got the environment and the agent
um interacting with the environment in a
particular way
it's being exposed to the sensory signal
but it doesn't know what the
outcome of the o was regenerated by
so it can only perceive itself and the
world around it
through only the o and it needs to make
inferences about what type of states
or um the true causes um
were um responsible for the particular
sensory input that it's being exposed to
and this is why um in active inference
when we formulate the problem we
formulate it as a partially observable
markov decision process
because um in this way we are able to
formulate a gender model that it defines
this
internal distribution over the internal
states that the agent
would use in order to infer the outcome
so he doesn't have access to the true
state
but it can make hypothesis or beliefs
about the states that could have given
ris
rise to a particular sense uh outcome
space that is being exposed to
um and using this the agent will make

inferences about the true state
using a process of reverse mapping
specifically bayesian model inversion
and to make this a little bit more
concrete what you can do is think about
the hidden states as locations or color
for example and the observation space
that the agent would be exposed to would
be

um for example the the velocity of the
movement
or a particular reward uh or like a
happy face that they're being exposed to
um okay so if we were to think about
this a little bit more formally so
what is the gentle model so as we um
described before a gentleman is a
partially observable
mdp within this active and friends
formulation

which rests on a simplified setting that
we're considering here where we only
have two random variables
the first one is o that we've discussed
and the second one
is s where s denotes um a random
variable representing hidden or latent
states and they exist within a finite
set of all possible
hidden states which is denoted by
capital s here

and this joint probability that we get
over o and
 s can be factorized into the likelihood
function

which is $p(o \text{ given } s)$ and then you
have the prior over the internal states
which is $p(s)$

so this gives you a very nice
formulation um
that we're going to use in a few
next couple of slides and i just wanted
to ask
are you able to see my mouse when i
highlight or when i
or is that not there i can see it yes oh
yeah perfect
so we know that for an agent to minimize
its surprise we would need to
marginalize out all the possible hidden
states that could have led to a given
outcome
and this can be achieved by using the
factorization that i just mentioned the
likelihood
and the prior um but the
the problem is that um this is not a
trivial task because the dimensionality
of the hidden states
um can be extremely large and if you're
considering additional random variables
that we're going to
introduce in a bit this becomes even
more problematic
um and that's why we use uh another
quantity a variational approximation of
this quantity
uh $p(\mathbf{z})$ which is more tractable and
allows us to estimate the quantities of
interest um so
so this will um be a natural step to
talk about the variation free energy
which is this variational approximation
of the quantity of
interest um so what is variational free
energy so variation free energy is

defined as the upper bound on surprise
so the first definition
sorry first definition that we
considered is derived using Jensen's
inequality and
is commonly known as negative evidence
lower bound in the variational inference
literature so we get this from the
equation 4 that we just saw
by introducing negative log on both
sides
and then um multiplying this term
by 1 which is essentially $q(s)$ over $q(s)$
of s so we're assuming that
 $q(s)$ cannot be equal to zero and
with that um we then apply Jensen's
inequality and we move the log inside
the function
and we end up with our expectation with
respect to t of s
for log of the joint over the
approximate or the variational
uh quantity of interest here and then we
take the negative inside and we can flip
it around
and we get our first
nice quantity of interest here where we
get
the the bound that we're interested in
and
in terms of Kullback-Leibler divergence between the approximate
and the joint that we have
so to make this a little bit more
concrete what we can do now is to
further manipulate the variational free
energy
summons um into uh the Kullback-Leibler divergence between the
approximate and the true posterior

minus the the log evidence that we the
 model evidence that we had
 and we can re arrange the last equation
 to really hone in on the connection
 between
 surprise and variational free energy um
 so
 if you remember that kl is a divergence
 which means that it cannot be less than
 zero so
 um it's always strictly uh greater than
 or equal to zero which means that
 when our approximate is equal to the
 true posterior
 we end up with the variation free energy
 equal to the model
 evidence which means that minimizing
 free energy is essentially equivalent to
 maximizing the model evidence
 um okay
 um we can rewrite the the previous
 equation that we had
 um equation 10 to express the
 variational free energy
 as a function of posterior beliefs in
 multiple different forms
 so i'm just going to focus on equation
 12 here which is the complexity
 minus accuracy so this is a trade-off uh
 that's normally used when
 the papers that essentially says um the
 complexity tam
 or complexity cost is essentially um
 your trade between your approximate um
 of s given p_i with respect to
 uh your p of s given um p_i and here p_i
 is just your policies and these can be
 regardless of hypothesis of

how uh the agent is going to act
but i'll come back to what policies
really entail but for now
just consider them as a term that allows
us to condition the free energy
on a sequence of trajectories of
interest
and the second term that we have is the
log probability
of o given s so the likelihood uh
with a with respect to the key of s
that gives you the accuracy so a simple
way to think about it is that
um this is just um um how accurate the
model is and this is some
regularization term that's a penalty
term to make sure that it's not
diverging too far away from our
initial priors
okay so this particular quantity
variational free energy
are there any questions at this point
about the variation for energy
not yet thank you okay perfect okay so
the variation for energy is giving us
this way of perceiving the environment
and addresses one part of the active
influence formulation which is making
inferences
about the given world that the agent is
interacting with at a given point in
time
however we have not actually accounted
for the
active part whereas like this particular
agent
that we have under the active inference
formulation can

take series of actions or interact with the environment in such a way that it affects that environment in the future um

so to motivate this a little bit further what we can think about is that not only do we want to minimize our variation free energy we also want to minimize a quantity called expected free energy which depends on anticipated observations about the future in the future or about the future and minimization of this

particular terms allows the agent to influence the future by taking particular actions in the present which are selected from a set of policies so i mentioned policies a few times now

so what are they um so policies can be defined as a sequence of actions at time τ

that enable an agent to transition between hidden states and τ here is essentially as a sequence of trajectories up to a particular horizon

uh τ which is um considering the total number of time steps that you are considering in a particular setup

and for us to properly define uh policy we need to introduce two additional random variables so the first one is an action that's conditioned

on τ which is denoted by a_τ here and this exists within a finite set of

all possible actions that the agents can take
and the second random variable that we introduce is uh
policy which is the π that we've discussed
and this exists within a finite set of all possible
uh policies um or sequence of actions in a sense of like the sequential policy optimization that we're interested in here so to make it a little bit more concrete χ
hair the random variable can be decomposed into
a series of actions over a particular time time horizon τ
so u_1 u_2 and u_m going up to u_{τ} would denote the action from
uh action at time point one action of time point two and so on
and the link is explicit when you consider that
if you consider a policy at a particular time point so τ then the action that you get would be that action
okay cool um so
i also wanted to highlight that this definition of policy is actually quite different to
or distinct to how it's considered in rl
which is
um when they say policy that means state action policies
so as i just mentioned an act of inference a policy is simply a sequence of choices for actions through time that is a sequential policy

and this is different to a state action policy in reinforcement learning which is mapping of states to actions so your rl policy which takes in account the action and the state is probability of your action given the state and under um our mdp formulation the definition of action oh sorry definition of policies in rl and active inference become exactly the same when we consider the setting where τ is equal to one so you're only considering one step ahead okay okay so um i'm going to move a little bit and consider uh the quantitative interest they expected free energy which is how do we even derive it so in order to derive it we first need to extend the variational free energy definition that we had before so a few slides ago and now make it dependent on both time so τ and policy and what we're doing essentially is taking that same equation and just decomposing it for previous sorry previous and current time step under a particular policy so that way this is why we have the conditioning and then we're decomposing it in a specific way um in equation 15 and then writing out the matrix formulation in equation 16. so we can come back to this if there are any questions

but the key thing to take away from the slide is that now we're including a functional dependency on time u_m for the variation for energy u_m and this is allowing us to now move to the expected free energy formulation u_m but the key thing to note here is that we're only considering time points u_m the previous time point and the present not the future at all so using the free energy equation before we can derive u_h the expected free energy u_m and what is the expected free energy then so expected free energy is u_m the free energy function of future trajectories g and it effectively values the evidence for plausible policies based on outcomes that have not been observed yet so that's the key thing so you're making inferences about the polls the set of future trajectories that you haven't observed u_m and there's a two heuristics that are introduced here in order to get to the g formulation that we see in equation 17. the first one is to include beliefs about future outcomes in the expectation that is we're supplementing expectation under the approximate posterior with the likelihood here which results in a predictive

distribution given by these first two
terms here
and the second one is that we're
implicitly or
i guess explicitly conditioning on the
joint probabilities of states
and observations in the gender model um
sorry in the gender model um dependent
on
the desired state of affairs as opposed
to a particular policy now so this
constrains the type of preferences that
the agent
would have and what's helpful with these
two
moves that we're making is that we can
now evaluate this quantity
before actually having the observations
and the second one
is that the minimization of g would
actually encourage policy
policies to be consistent with uh the
desired state of affairs that the agent
expects itself to be in
um i'm just going to briefly mention
that this is not the only way to derive
expected free energy and there's been
some work um that's looked at other
formulations
uh including work by carl um where
the formulation of the expected free
energy can be decomposed into different
um structures so if anyone's interested
in that we can go through it later
um but this free energy expected free
energy that
i just introduced can be decomposed in
certain ways um so equation 20 and 21

give

two different decompositions of the
first one being the epistemic and the
extrinsic value trader and the second
one being the expected and the
uh cost and ambiguity term so if we just
consider the

um the first equation we can say that if
we're minimizing this equation
then we're capturing this imperative to
maximize

the information gain that you would have
from observing the environment
um about particular hidden states
while maximizing the expected value
which is scored by the
log preferences or the extrinsic value
here so

this particular formulation actually
gives us a very clear trade-off between
the

um the first component which is the
epistemic value that promotes
curious behavior so that's what you want
with exploration encouraged as the agent
seeks out

these new states that minimize
uncertainty about the environment
and the latter bit is more pragmatic and
it encourages

exploitative behavior through um this
understanding of the type of
uh policies that the agent would prefer
to reach

um in other words like this expected
free energy formulation
and that we're seeing in equation 20 is
essentially treating

exploration and exploitation as two different ways of tackling the same problem

so minimizing uncertainty that mentioned at the start of the presentation

okay um we can also think about the second equation

here which is just offering us an alternative perspective on the free expected frequency which is an agent wishes to minimize the ambiguity and the degree to which outcomes under a particular policy deviate from prior preferences thus ambiguity here is the expectation of the conditional entropy or the uncertainty about outcomes under the current policy in this particular setting

low entropy would suggest that outcomes are quite salient and uniquely informative about the hidden states so for example the visual cues that you might see if the room is actually completely tough

in comparison to if it's quite dark you're not going to make out anything important from that in addition um the agent will like to pursue policy dependent

outcomes that resemble its preferred outcomes

so it's the donated by c here and this is achieved when the care divergence between the predicted and the preferred outcomes is minimized by a particular policy

okay um and these prior beliefs about the future

outcomes equip the agent with
goal-directed behavior
which is one of the i guess the
instances that
is really important in active inference
um okay so once we have the expected
free energy we can derive
um the the policies um so
um and this is realized by deriving the
probability of any policy
by applying a soft max function over the
expected free energy
and this sort of illustrates the
self-evidencing behavior of active
inference
because any sort of policy or action
sequence that results in lower expected
free energy are more likely
and intuitively this would uh make
sense because the expected free energy
is sort of
encapsulating all the types of things
that you want to include
or consider when you're interacting with
the world so you want to explore you
want to exploit but you want to have
a balance of that um and then
when you're selecting your policy um
it's just a matter of determining the
set of
actions which get you closest to this
particular core
um and this can be defined by an
attractive state that's defined by your
c matrix that we
described before um and if you don't
have that then
it's just random exploration that you

would get

um sometimes you can also include a temperature parameter β here um and by having a hyper prime on this you introduce an

additional complexity cost into the expected free energy

uh formulation which allows you to account for um how flat or how

confident or precise you want your preferences

to be over the policy space

um the key thing to note is that um for a sake of simplicity i'm not going to go through a lot of the the details about how these are optimized

but you can do that in multiple different ways for example in active inference

we can optimize expectation about the hidden states of interest the policies the precisions that are inferred and then we can also optimize the model parameters through um through the learning procedures involved

um but those sort of differ depending on the setup you're looking at for example if you're using variation base you would just iterate

this uh these functions or objective functions until convergence or an active inference um you do a gradient ascend

um to find the sufficient statistics of interest again this depends on

exactly which formulation and setup
you're looking at
but the key thing to note here is that
um that there's
three particular aspects to the active
inference algorithm that are useful and
can be
uh taken from this particular uh
framework
and applied to uh other settings
um so i'm just gonna reiterate and
summarize them briefly
so we first have the gender model that's
crucial so in order for an agent to
interact and minimize its surprise it
needs a gentle model of the world
and that's described as simply i'm not
i'm not including any of the model
parameters here but you can have
your outcomes your states and your
policies and this
um these are decomposed into um
yes sorry there's a arrow with the
brackets here but
these are decomposed into your prior
your likelihood and your transition
function
and then you set once you have this
genital model the objective the agent
is to fit the model to sample
observations to reduce the price
and that is through variational free
energy optimization so this particular
trade-off that we have between
the complexity and the accuracy cost and
then
the second oh sorry the last part of
this algorithm is to plan

so select actions that minimize
uncertainty
that um that is the expected free energy
and the way you do that is by having a
soft max
over that this uh g negative of g the
quantity we have here
and then sampling from that in order to
select the next specs
best action okay
so that's a quick and deep dive into
a massive amount of active inference
literature but i just wanted to
highlight that
those are the three core ingredients
that if you are interested in
implementing these algorithms yourself
okay so now i'm just going to switch
gears a little bit and walk through
comparisons with
reinforcement learning so in our work we
considered a modified version of the
open ai
gyms frozen lake environment so frozen
lake has a grid-like structure
with four distinct patches so it has a
starting point which is
s um so we can see it here um apologies
but it's super
tiny so s is here and you've got the
frozen
surface which is f so again i don't
think i can differentiate because i'm
just moving my mouth
um i'm zooming in they can see it
perfect so you've got the frozen f and
then you've got the whole
and then lastly you have the goal so g

here
where the first b is located and all
patches in this particular setup
are safe except for hall where if the
agent goes
to h it gets a negative reward um
the agent starts each episode at the
first
um position which is the starting
position and from there it needs to
reach the frisbee location
in um in the least amount of steps
possible
and the way you can do that is by
performing uh four different types of
actions so either going
left right down or up and
the agent is allowed to carry on moving
through the frozen lake
uh with multiple revisits so you can go
back to the starting position having
like on in other places
but each episode will end when either
reaches the
hall or the goal location and
these locations differ depending on the
setup that we'll
um have in our simulations so
in one set up um the position of the
whole is
eight and the goal is six and another
set up the position of the whole
is six and um
the goal is eight and the objective is
as i said to reach the goal
in ideally as few steps as possible
while avoiding the whole because that
would end the episode

uh if it reaches the goal it gets a positive reward of 100 and negative otherwise um the key thing to note here is that this query metric actually allows us a way to compare the active inference algorithms to reinforcement learning algorithms but it's not really important for the active inference algorithm to um have the reward function as a get-go because it can still move around and using just the information game tab so not having the extrinsic value component um and that that's quite interesting because we'll see the ramifications of that in our assimilations and for this particular setup we limited the maximum number of time steps for each episode to 15. okay so what i'm going to do is first talk through the genital model that we use for the active inference formulation um so here what you're seeing on the slide is a graphical representation of the active and for instance model so this model contains four action states so right down up and left and these control the ability to transition between hidden states uh location fact so for example if you are in position one and you take the action right then you'll end up in position two or if you're in location

five and you take um the up action
you'll end up in position two
as well um
in this particular setting both
positions
six and eight are absorbing states
because if you'll remember
once the agent goes to that location
they're not able to move
out so that's when the episode ends um
and if an agent makes an improbable move
in this particular maze for example if
it tries
to go from position one to left it will
just stay in that location it won't move
in this particular um gender model so
i'm just looking at the hidden states
now
we have a chronic um transfer product
between the two factors so we have
location and context here um
the context cannot be changed by the
agent
that we have um because this is
something that's determined by the
environment and
this determines where the goal of the
whole locations are whereas the location
is something that the agent has control
over and that's where we have the action
states
over this um so with the context we have
two contexts
the first one is where the goal is in
your location eight and the hallways in
location six
and the second context is where the goal
is in location

six and the hall is in location eight
um at each time point the agent will
observe uh two outcomes one would be its
own position in this particular maze
and the second would be the score that
the agent would get
um the likelihood for the grid position
is entirely determined by the location
of the agent
and the score um is determined by
both the location and the context in
place so
if the agent is in location six
and it's in context two then it will
receive a positive reward
otherwise it will receive a negative
or mutual reward
depending on where it is um
based on um trying to make that
comparison with reinforcement learning
what we're doing here is we
are introducing uh prime preferences
where
the agent has plus four um for
um positive reward negative four four
sorry minus four for negative reward and
otherwise and it
at the first stage it expects itself to
be in the
first location
so we compared um this particular
gender model and the active inference
agent to
two reinforcement learning algorithms so
the first one was the q learning using
epsilon greedy exploration
and the second one was the bayesian
model based reinforcement learning

algorithm using standard thompson
sampling
and thompson sampling is appropriate
procedure here because it entails the
optimization of dual objectives
reward maximization and information gain
and this is achieved by having this
distribution over a particular function
that we prioritize by
a prior just by having a prior
distribution over it that we sample from
okay um
so for the two q learning algorithms we um
have two epsilon greedy uh parameters
so one where it's fixed exploration set
to 0.1
and then another one where we have
decaying exploration that starts from
one and decays down to zero
so first we assessed um this um how the
agents interacted in a stationary
setting where the
reward wasn't changing uh such that the
goal location was always at six and the
whole location was
always that age and then we evaluate the
performance of the agents um the key
thing to take away from here is that
both the
bayesian rl and the active inference
agents are able to
quickly learn um where the reward
location is and just maximize it out
and this is um this is performance is
consistent denoted by the really tight
confidence bounds that we see
in comparison the q learning agents are
um so for the one where we have uh fixed

exploration

it's fairly um fairly good and is able

to land where the reward

is located but there is um some

deviation denoted by that

ten percent of selecting a random action

um whereas the q learning where we have

epsilon is equal to one to gain to zero

um the performance isn't the greatest

and for the null model the active

inference where there is no reward here

the agent randomly goes to the hall and

randomly goes to the goal

50 of the time but the key thing to note

is that apart from the non-model

all the models are doing fairly well and

seem to be

performing okay within the stationary

setting

so the next thing no could i just make a

point about these experiments as well

just because

they also do veer away slightly from

like the traditional rl like

which is yeah great so basically

like usually in rl what happens is you

kind of have this ambiguation between

like training time and test time

performance and you usually especially

in something like q learning

like you just hit the max over the q

function or you have a policy

that tries to do like some search over

the q function

and takes that max like given a state um

but there's kind of an artificial

distinction which is

like obviously as you're acquiring data

you're making mistakes in the environment you're interacting with the real environment so in order to kind of make that fair comparison between here and active inference where you don't really have this distinction between like training time and test time it's all just interaction the that's the reason why the q learning agent especially when epsilon is fixed to say 0.1 never achieves the optimal policy simply because we're also with 0.1 probability taking a random action so we're not making this distinction like you sometimes would with normal rl between train and test time we're all saying train and test some are basically the same thing so however you're choosing to interact with the world is how you should be assessed so that's that's why initially looking at these girls you might be like hang on like why isn't q learning solving this but um yeah that's just to clear something up if you're more familiar with say some of the like dpri uh experimental procedure perfect thank you okay um and then um just following on from that so we changed the environment a little bit to make it a little bit more difficult to see whether

um the bayesian and the active inference
agents might struggle
when we start having the relocation
change after every few
episodes so specifically we swapped the
goal and the whole location
at um at time point 21
at time 121 141
251 and 451 so you can see that
in these figures where the line the gray
lines are shown so these points
they're all location flipped so like the
stationary setting for the first
20 trials all the asians seem to be
doing
as you would expect so both the bayesian
rl and the active inference agents are
doing
fairly okay uh the q landing with
um the fixed exploration sector 0.1 is
doing fairly okay
as we saw before and the acute learning
with the
decaying exploration is is doing as it
was doing before
but when you flip it around um such that
the goal locations
change um what you notice is that with
the
bayesian rl agent the the amount of
reward or the score it gets
is quite low and then we we see a phase
where
it then transitions and becomes um to
the optimal policy again
in comparison to the active inference
agent where
it instantly after the first trial of

doing the incorrect is able to switch
over to the active and sorry the
appropriate policy
and the reason for that is for these uh
rl settings where we're considering it
as a learning problem
you need to first do reverse and
learning of where the reward location
is and then we learn the new reward
location so you're seeing that for
q learning uh for both the different
epsilon greedy parameterizations and
also for the bayesian rl and we're
seeing that consistently
whereas for the active inference agent
because we're treating it as a planning
as an inference problem where the
posteriors from the previous state are
moved over to the prize and moved over
as priors
and the agent is able to instantly
um realize that the current policy that
was following
the this time step is an appropriateness
which is its policy to the next
um to the the other one again
um the null model as expected doesn't
really do much because it's just
exploring doesn't really care
where the reward or the whole location
are
okay um phil do you want to add
something to this or
no i think the same point applies to
this one
yep definitely okay um
so just to wrap up with these two
comparisons and

i with the non welder stationary setting
um all types of agents um including the
bayesian rl and the q-learning would be
reasonable frameworks to use whereas
with the uh with a non-stationary
stochastic setting
having an active inference agent might
be appropriate way of handling changing
dynamics
um but the key caveat with that is that
you can
introduce a lot more additional
complexity into the bayesian rl or the q
learning or the rl framework in general
to allow
uh for a way to handle uncertainty but
it wouldn't be
a natural way of adding it you would
have to sort of uh augment the
function or the algorithm in particular
ways to justify it
um okay so so once we have done this
comparison we were interested in is um
why would you even want to use active
inference um
when um you don't have an understanding
about the world
or you don't have sort of a preference
over the type of things that could be
done because as we saw the active
inference model the null model is just
exploring it's not really doing much
um and that's brings us to one of the
initial points that i introduced
at the start of the presentation which
is that within an active influence
framework we don't really care about
having a reward function

we can learn we can learn that based on
some interaction with the environment
um so for that um
what we did was we carried out a few
different simulations to see how the
active inference agent can select
different types of policies in the
absence of prior preferences
and for this we we did three different
experiments
um where we allowed either or the
likelihood
and all the outcome preferences to be
learned
over time and saw how the agent
interacted when
this learning over the different um
preferences uh took place and the the
way we did it was using
uh the conjugacy models where uh because
these are discrete state models we have
categorical distributions over our model
parameters
and we're introducing directional
distributions on top
as a hybrid prior and learning those
hyper prior
um we can go into the details of this if
anyone has any questions but
just for simplicity um just assume that
all the dirichlet distributions uh were
set to flat completely flat and
the agents were all given an opportunity
to learn
based on different interactions they had
so the first set of
experiments like the simulations that we
ran were to understand how

um the agent would reduces uncertainty
about the
environment that is interacting with so
if it doesn't know the
the the the frozen lake
how would it interact or explore that
lake
um and what we see is that um
in this particular instance the agent um
was just interested in exploring it
didn't particularly care about
the where the goal or the whole location
was
um and this is highlighted in
a series of different exploration um
trajectories that we see so the first
one was where the agent falls in the
hall
um and then another one where it goes
around
and ends up at the second in the second
episode at the goal location
and in somewhere it just ends the
episode by
just going back and forth so this is
just floor exploration
nothing more to hear um
the next um set of analysis or
simulations that we ran was to see what
would happen if the agent knew about the
world but had no preferences of the type
of outcomes they
expected itself to be in and for this um
we ran multiple different simulations
and saw that in the absence of
um any sort of preferences
um the whole could actually become
really attractive

if it's encountered first so we see that
in the
first figure where the agent learns to
prefer to hide in halls
and in the second type of trial that we
saw was where the agent
uh exhibited the preference to actually
go to the goal location so this is
entirely dependent on the
instantiation or the type of stimulus
that the agent is exposed to
initially that determines the type of
preferences it would
learn to have
okay and then the last set of
simulations that we ran was just to
check what would happen when we
interacted with the
epistemic imperatives actually resolve
uncertainty about the
environment that the agent was
interacting with specifically the
likelihood mapping between the outcomes
given the states and
the uncertainty about the desired state
of affairs that the agent expected
herself to be in
and what we saw was that um if we
allowed um a sufficient number of trials
to pass the agent
in this particular setting learn to
prefer to ride sorry hiding holes
after x number of episodes but it had a
very distinct
um preference over the type of outcomes
it
expected itself to be in depending on
the particular

uh a sorry particular time point
um so that brings me to the last
simulation so i'm just going to wrap up
which is um in active inference
i started the presentation was saying
it's a particular algorithm that gives
us
very nice um things to consider when
we're
operating in a bayesian or belief based
setting which is
firstly the principal account of
epistemic exploration and intrinsic
motivation that we get from the
particular expected free energy
decomposition that we went through the
second thing
that i wanted to highlight was that um
under an active inference setting we
don't have to explicitly
specify a reward function which we saw
in the last
uh the second set of simulations where
the agent can also learn its own reward
and prefer to become something that is
quite counter-intuitive
from an rl setting where the signal from
the environment is saying something is
bad but the
agent's internal motivations of
preference allow it to then
do something that's quite uh at odds
with what the environment is expecting
it to do
and lastly because of this belief um
invasion setting uncertainty is a
natural part of the belief
updating um so within the

stationary settings active inference
agents perform as well as reinforcement
learning agents
however in non-stationary settings they
outperformed due to their ability to
carry out this planning as
inference and that's i think that holds
um this is not a conclusive statement
but
it's a nice way to start thinking about
how
um if you're scaling up active inference
agents to interact on the same type of
environments
as reinforcement learning agents that
might be
a little bit hard to resolve because
there is some
non-stationarity or some weird
fluctuations happening in the
environment
and active inference agents could
potentially perform quite nicely if we
have this planning as inference set up
um and that brings me to the end of the
presentation so i just want to thank
everyone who was involved with this work
so
called tom and phil and everyone who's
helped me think through these
interesting ideas that i presented
so and everyone also for listening thank
you
thank you awesome talk
so you can maybe unshare and we can ask
a few questions
and if anyone who's watching live would
like to ask questions they're

more than welcome so maybe i'll start
with just um
a general point that it was awesome to
see how clearly you differentiated
between
the reinforcement learning and the
active inference paradigm
and just one question i had was you
mentioned that
in the case of a time step of one a time
horizon of one
there was like an equivalence between
the reinforcement learning approach and
the active inference approach
now people are using reinforcement
learning to do
planning for the future amidst
uncertainty
so how are they accomplishing those
kinds of planning
and where are some situations where
active inference
might be able to step into those um
settings and
potentially do better i think that's
already a question
a good question and feel free for you to
jump in if i miss something but um
so within the url setting from what i
know if they are considering
temporal horizons greater than one then
they have hierarchical rl
or they have options where they're
considering trajectories that are
greater than
one that allow them to consider a whole
sequence of play
um before they like um that's rolled out

if they select a particular policy
instead of like that actually
one single action to state mapping um
but i i'm not i haven't really worked
with options that much maybe
phil do you know um yeah so options are
like one way to
achieve this kind of multi-step sort of
um like i guess oversized almost like
this inductive bias right that
you will like your choices will be kind
of contiguous blocks of like
more than one step whereas a kind of
policy kind of it's just like i'm going
to take one step and then i'll have this
kind of
next state i exist in um but i think
what's important to disambiguate is
actually you can do sort of planning the
model style things because
you can view active inference as a sort
of model based like
in the kind of you know rl paradigm and
indeed like within model based
like you can have
um like planning methods similar to what
is done
in active inference however i guess like
the key distinction here is
when you do planning in model based rl
you're
simply just trying to take you tend to
take some sort of
suit like kind of mini evolutionary
method usually something like
the cross entropy method and what you're
doing is this kind of search over all
actions that you take

at a certain point in time that
maximizes your kind of reward
over say a horizon of say 20 steps um
and you can do some clever stuff you can
terminate with a q value function
um but fundamentally like it doesn't
take you away from this sort of
definitional difference which is
in active inference you're doing this
sort of planning over actions even
though they might seem superficially
similar like kind of
basically like drawing lots of actions
and seeing which ones maximize some sort
of utility
in the case of active inference all the
useful kind of information and the
desired data about um
like exploration and it's and um
exploitation are wrapped up in that
maximization whereas in rl it's kind of
it's a bit harder to understand um like
in the classical sense we're just trying
to maximize reward but um
you can have heuristics where you say oh
but maybe i want to also maximize some
notion of
model uncertainty and you know it kind
of gets a bit
um more difficult to naturally integrate
all these approaches into the same thing
um because like in the example of active
inference because everything's a
distribution
you just put a hyper prior over if you
want to you just integrate it out if you
don't want to deal with
kind of tuning it so yeah that's kind of

where i feel like
there is this kind of difference
cool and what what kinds of settings
do you think that those types of of uh
action as inference and planning as
inference could be
utilized like what kind of data sets or
questions or contexts
are people currently using another type
of method but then you're excited to see
active inference play a role
i think mostly open-ended problems where
you don't really have a reward function
i think because
the way the rl agent is learning how to
interact with the environment is through
a reward function
um so anything where you don't really
have that or you have
um an environment that that is changing
um but i know that there's like a whole
host of
uh people within the rl community
working on like
intrinsic motivation or internal
motivation
so those sort of things do overlap with
the
active influence formulation um
but particular paradigms i think for me
the more interesting aspect of active
influence comes from the fact
when you start thinking about um
biological agents and if you're modeling
a patient
or someone with schizophrenia you can
with this bayesian framework you can
change um the priors to try and see

how the person is interacting
where i guess it's it could be that
their policy
like the way they're valuing their
policies are broken or it just could be
other parts that are different but
i think within the rl setting if you
are stepping outside the um
the standard the game mode right so like
magic or
gym environments and start going into
ones which are more open-ended and you
don't have any rewards then
active influence could be potentially
useful but
i i'm a little bit hesitant to say
whether it will be
better because if you start
augmenting bayesian rl with all sorts of
interesting components
then to a certain extent it will be
active in front scaled up um
which which is potentially a contentious
point
um but i i think it it depends on
exactly what you were incorporating and
that's why for this particular
presentation and our work we were quite
careful in defining what
reinforcement learning meant which is
that you have to have this reward
function in play and you want to
maximize this reward function
and anything you include into the rl
framework has to have
that uh that like the object that
objective has to be there but if you
if you sort of bypass them say okay i'm

just going to add in all sorts of
interesting components
to make the algorithm or the way um the
agent
interacts with the environment um
i i guess as similar to a active
inference framework
or even perhaps even better then you
don't
that fine distinction between where rl
is better or where active influence
is better sort of it's not really there
for me
um because i think both communities from
my
perspective are working from similar
things which is sequential decision
making
um and with our work is mostly
with uh sequential decision making in
the face of uncertainty
whereas some of our work my rl work
might not be
focused on that so i think it's when you
start
drawing the boundaries it becomes a
little bit hazy where like
things are separate or not and i think i
went up on a little tangent
um but phil do you want to add
anything to that in terms of the
environments and the paradigms that
might be
useful yeah i think if
one of your aims is say you're given an
environment that you just have no prior
knowledge about
how you ought to behave you could argue

that you could deploy an rl agent that
uses some sort of curiosity or epistemic
uncertainty reduction mechanism but
i mean i know i know there is like a
tiny bit of work about learning priors
over reward functions and learning those
but

i'm not hugely like aware but um
i think what's important to understand
is in the limit of like exploring the
entire environment
your epistemic uncertainty is going to
go to zero right like you um
you will have observed everything and
then it's unclear what your rl agent is
going to do at that point especially you
say you have a

deep neural network that parameterizes
um how you take actions

whereas in the case and i think what
this paper was really interesting
for me to see like when we ran these
experiments was

actually you just put a prior over your
prior preferences and eventually you
learn

a mode of behavior it may not be optimal
but your agent eventually

learns to adopt a behavior that is
self-fulfilling because it reduces
epidemic uncertainty

and then all that's left is says well i
think these are kind of

useful behaviors or at least these are
behaviors for me to do in the world and
eventually

you get quite a repetitive behavior and
it might be a more kind of accurate

simulation of
in the absence of any information like
how something intelligent might actually
behave
in a world whereas in the rl paradigm
it's less clear you know once you reduce
all that epistemic and certainly about
where the value is
and you haven't found any like what is
your agent really doing at that point
but i think my problem at the moment
with like the acting from
formulation is that this works on like a
discrete state formulation right we saw
really nice results in that setting
but i think if you scale it up active
inference agents are going to have the
same
issues if you're using like amortized
inference or something to actually
uh approximate your likelihood
transition functions which means that
you might not get these nice properties
that we're seeing at the
this small scale um so scaling it up is
like a
it's a interesting and open-ended
problem at the moment scaling up in the
right way that you can
include like these conjugating models
that we have
in um the discrete state formulation and
that's how we're
like do the last set of simulations
where we introduce the conjugate priors
to do the
learning over the the prior preferences
but also the likelihood

um so it becomes a little bit hazy how
to learn

like i have a hyper prior over an entire
neural network if you're scaling it up
that way

um so they must like a little work needs
to be done that area

and in order to actually show that it's
appropriate

um that you can have these interesting
components then you need to
start thinking about how you would
include these hyper priors

uh because it is not reasonable to say
that you're gonna have a hybrid prior
over the the parameter space such
like the the beta hyper so that hyper
prior over the gamma that that would be
sufficient in those settings

um hyper prior for the way the agent is
selecting his actions

it has to be over the model parameters
and if

scaling up means that you're losing that
nice way to disentangle the particular
model parameters then it becomes very
uncertain i

i don't know it's it's an open-ended
problem for me

cool yeah i think high high-dimensional
problems

still represent something that um is
relatively difficult especially

just due to the kind of you know you
know the further we stray from bays
like um the less principle it becomes
and um you know

it's like it's it's quite a fine line

very interesting about how whether
within the rl
or active inference paradigm there's
sort of the
sparse skeleton the bare bones at the
core
and then sometimes these other layers
or tweaks are needed
and pretty interesting to learn about
and also what you had said earlier
about how the challenge is planning for
sequential action
when you're in feedback whether just by
moving around an environment so your
local environment changes or you're
playing a game
with a board game is going to change or
you're trading on a market
sequential action you can't just plan
steps one through 100
without at least thinking about some
what-ifs
and then only having access to limited
observational data and planning amidst
fundamental
uncertainty so i think a lot of those
points of contact with the motivations
of reinforcement learning and machine
learning will
maybe bring some more light into active
inference and
um push some of those frontiers you just
mentioned
so i have a question and anyone else in
the chat can ask a question too um
let's say somebody is wanting to learn
about
this and they're actually in a lucky

beginner's mind perspective
because they might not have been enticed
by learning reinforcement learning but
they've gotten curious
in active inference and excited by your
presentation
and so what what kinds of computer
languages or
skills might they want to learn or what
kinds of approaches or mindsets would be
helpful
if somebody let's say weren't coming
from a
classical machine learning perspective
and learning active inference but rather
kind of upskilling into active inference
what would you recommend either of you
to them
bill do you want to go first with that
yeah i mean i think
it's an interesting way to view a kind
of this kind of potential person because
i kind of felt like
i was somewhat like this person like way
back when
like nora and i was starting to have
these conversations about active
inference um
and i like the the this paper started
off basically as a
tutorial i was writing having spent two
three months kind of
in the evenings reading about trying to
like sift through
the act of inference literature and i
think um you know i think it's fair to
say that at times it's unapologetically
dense

and quite difficult to read so you know
without trying to you know
self-endorse but i do think reading this
manuscript in particular like the whole
aim
philosophy when we like when we started
writing this was to
really understand like what is what is
happening here like what is this kind of
expected free energy quantity like why
do we care about it
so i know from a kind of theoretical
perspective i think this is a very
lucid presentation of the concept so at
least
you can get some sort of intuition as to
what's happening um as for the
kind of coding side i can't really speak
to that i'm sure i'm sure nor
kind of has has worked it works with it
quite a bit
um i i was going to say it sort of
depends on
what the primary objective of the person
is is it for a
sort of get an understanding of
the the high-level conceptual ideas or
treat it as an algorithm because if
you're going coming from a free energy
principle too active inference is a
different story or if you if you're
taking active inferences of siloed
a specific kind of sequential
decision-making
uh scheme right um so depending on that
is
sort of differs uh but as phil was
saying i think this paper definitely

is really nice in the sense like it does
try and define all the different
concepts and goes through
the different formulations maybe not in
as much detail um with the assumptions
in play it gives you
the the layout of how you might be able
to derive it
um but certain things like what
what the approximate density even really
entails those are very uh difficult
questions
where you would have to like dive into
variational inference literature to
understand so
i guess from that perspective someone
who's coming into the field
should spend some time thinking about
variational inference and how that
really
ties back to the act of influence
formulation
because the perception part of active
inference is
in most instances exactly the same as
variational
uh inference literature and optimizing
uh the model evidence
uh or the maximizing the evidence law
abound
um this is the second thing that i was
going to add
is um that the paper
with lance lot de costa is really good
for someone who wants to drill down
into deriving everything themselves
uh sometimes it's super technical so
this

art that the paper that i walked through
today uh the one with phil and tom and
carl
is it's a nice introduction for someone
who's not familiar with the mathematics
and just wants to
get a layman's summary uh whereas this
the paper the the paper where lance's
first author gives like the
detailed derivations and a lot of the
assumptions
in place so that's from like
understanding the theory part
um from the coding perspective it
entirely depends what the end objective
is so if they want
someone wants to work with discrete
state formulations
then the matlab code carl has written
and it's years of work with lots of
um nice simulations and examples that
you can use
and also our code is online and you can
access it
so there's a link in the paper um in
the software um section that gives where
exactly
the code is and you can look through
that and see how the simulations were
done
if someone is interested in um
more i guess um high-dimensional
formulations of active inference that
um there's some recent work with uh
zephyr zephyrus um so zeff fontes
as first author uh where we've got a
nice again we have a
git reaper for that work as well which

uh
gives uh a breakdown of how you would
implement a simple
active inference agent using uh very
short encoders and a simple
transition network so there's lots of
like different
areas but for someone starting out they
sort of have to understand whether they
want to focus on the
theoretical side or whether they want to
focus on the implementation side
theoretical side
would be drilling down into the
variational influence and the maths
behind it and if they want to focus on
the
the coding aspect then they want to
figure out whether it's continuous or
discrete state formulations are
interested interested in and then
sort of break down into if it's
continuous then
most of it would either be writing the
the coordinates of motions themselves or
they would have to use like some sort of
like um um a neural network to
approximate that continuous distribution
of interest
out or the discrete state which carl's
written out
um and if if they have questions about
the discrete state
um i'm i'm happy to take emails as well
so
if anyone has or the the continuous
state state
um state space as well thanks for the

distinction and um it's such a large difference between the matlab code which we got to walk through with ryan smith and christopher white which is doing matrix multiplication and such and then here comes the neural networks and it's offering sounds like new opportunities with high dimensionality and continuous variables but also a lot of new challenges so what is the essence that's shared by the matrix form and by this more machine learning style because for some people um it might be splitting a hair quite literally the difference between two computer languages when they're thinking about active inference from an ecological psychology or in an active philosophical or an embodied performance perspective all backgrounds that converge on active inference and so to somebody outside the the strand of hair what is it that we can really distill that is core active inference and i wrote down a few things so that you had said what are those core pieces that allow us to dive into the matrix mode with matlab or into the neural network mode with like python

um so i think it comes down to my
summary slide the way i think about it
so the active inference the core
ingredients are formulating the gender
model

and here formulating the general model
is the parameterization of the general
model so either you can use the
discrete stage categorical distributions
or you can use
a more continuous state formulation and
again

the neural network formulation is a
specific instantiation of that
so that would be one way of sort of
differentiating
that the second one is the
optimization of the objective functions
in play

so in the the matlab code
we're doing gradients to send using mean
field message passing algorithm so
specific formulation that's been
introduced in a couple of papers
and i specifically didn't walk through
that or

uh you're doing back propagation to
actually

calculate the θ or λ distributions
and then you're just

solving those distributions that you
have so it sort of depends on

which formulation you're um

like um it depends on how you optimize

those objectives either you're like

taking the implicit forward model

or you're taking an explicit gender

model um

and one thing i forgot to mention is
that um
alex shams and connor hines they've been
working on a discrete state formulation
of um active inference in python
which might be in of interest for people
who want to like focus on one specific
language
that can do like the more high end or
high dimensional stuff
and the discrete state formulation that
carl has um
i i know they're also looking for people
to work on the code base if anyone's
interested so it's called
infer activity i think um but that's on
github as well if anyone's interested
cool um any sort of last
thoughts or comments from either of you
um no i think i'm okay
so what do you reckon
um
i think we're kind of noticing a bit of
a
um an increase in interest in active
inference like
just to give you an example like maybe
areas where
even active inference would have been
considered before like robotics
um you're now beginning to see more and
more of it so i think if you do want to
get involved in it now is a particularly
good time
great call philip and i'll
um just re-recommend the excellent paper
that we're discussing it's in the
description of this video really

appreciate both of you
for joining you're always welcome to
come on to be
speaking about a paper you've authored
or not but
again thanks a lot uh to nora and
phillip and i hope to see you again on a
future
active inference stream thank you
thank you for inviting us bye-bye see
you daniel
peace bye
awesome stop the stream great
conversation thanks a lot to philip and
noor really